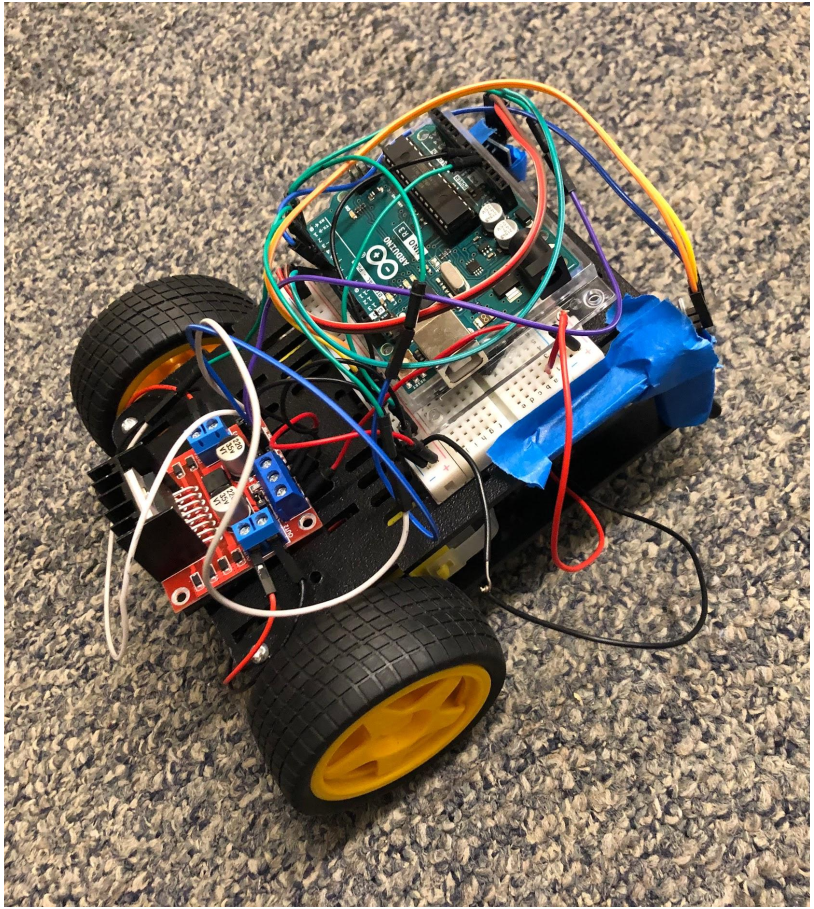
LINE FOLLOWER

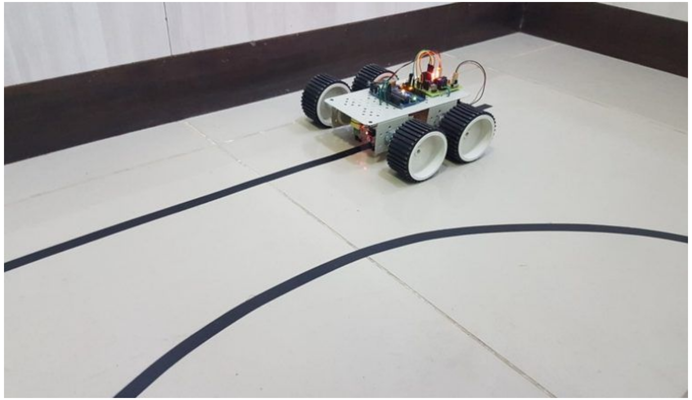
Marek Dec, Peijing Xu, Curtis Sung

Agenda

- Problem identification
- Customer needs
- Specifications
- Design process
- Testing methods
- Results
- Closing remarks



PROBLEM IDENTIFICATION



Automate a car to follow a line

Uses in

- ❖ Hands on experience for engineering education
- ❖ Transportation for warehouses
- ❖ Delivery services in hospitals and restaurants

CUSTOMER PRIORITIZATION

No.	Customer Needs	Importance
1	**Work reliably, especially on the day of the competition**	**5**
2	**Satisfy basic requirements for the competition**	**5**
3	Adaptable to varying competitive test conditions	4
4	Easy to assemble, able to be assembled under 10 minutes	4
5	Inexpensive components, under 30 dollars	4
6	Can survive a 1-foot drop in a backpack	3
7	Look cool with as many features as possible.	1

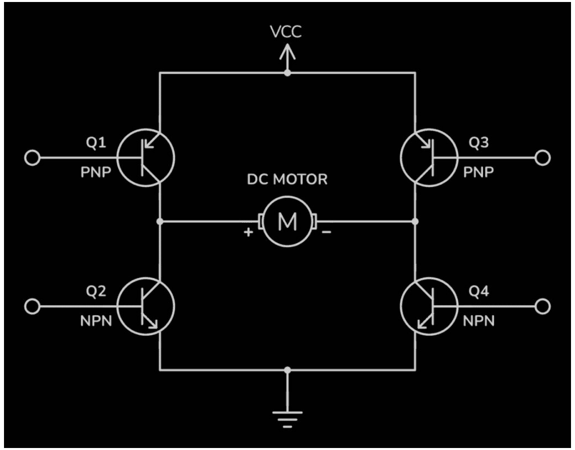
PERFORMANCE SPECIFICATIONS

Important performance specifications:

Customer Needs	Technical Specifications		
	Metric	Target Values	Actual Values
User Comfort	Weight (kg)	235 g	275 g
Size	H x W x L (cm)	10 cm x 20 cm x 20 cm	6.5 cm x 14.5 cm x 13.5 cm
Product Efficiency	Max Speed (cm/s)	30 cm/s	25 cm/s
Product Efficiency	Turn Radius (cm)	10 cm	12.7 cm

ALTERNATIVE SYSTEM CONCEPTS

- Build transistor H-bridge instead of the motor driver
  - Harder to implement
  - Cheaper and available materials



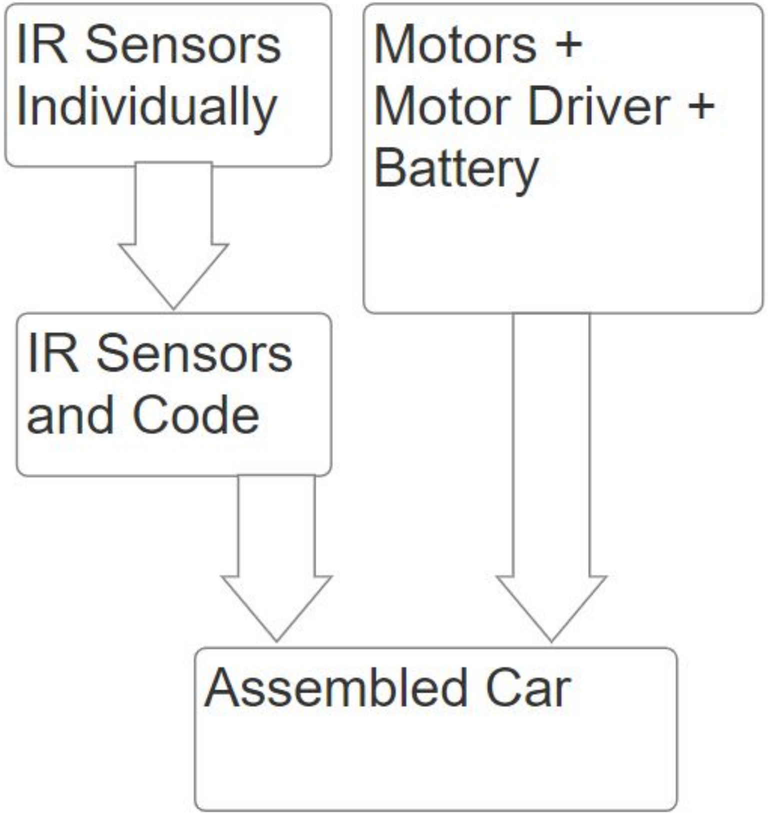
- Use camera instead of sensors
  - More complex
  - Likely slow to react
  - Higher potential for path detection

TESTING & ANALYSIS

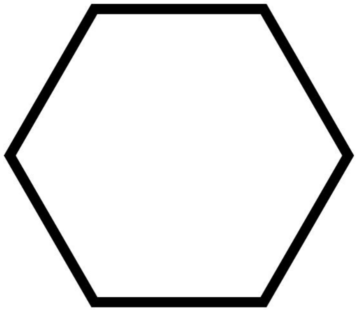
- IR sensors
  - Distance to paper
  - Different surfaces

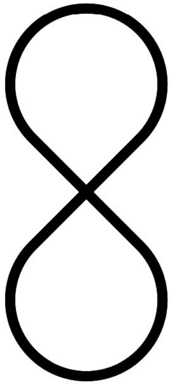
- Assembled car on test paths
  - Created custom paths
  - IED workshop paths

Used terminal prints to debug code



TESTING & ANALYSIS



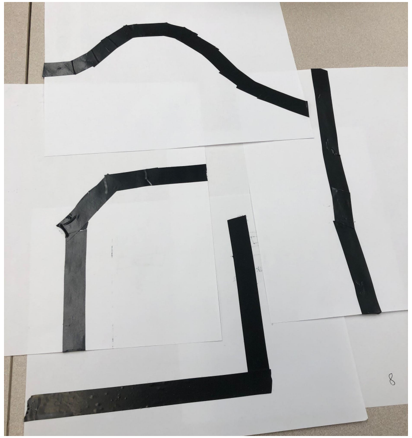


IR Sensors

- ❖ Worked better closer to the ground
- ❖ Adjusted sensitivity
- ❖ Detected all black lines we tried

Assembled car

- ❖ Passed all custom test paths
- ❖ Completed hexagon
- ❖ Struggled with the infinity
  - ➢ Intersections confused the car
  - ➢ Turns were too tight



TECHNICAL CONCLUSION

Contest result matched expectations

- ❖ Completed the basic course
- ❖ Intersection caused failure in the intermediate course

Possible improvements

- ❖ Add more sensors
- ❖ Put sensors closer together
- ❖ Screwing parts on instead of tape

LESSONS LEARNED

- Organize a form of communication earlier
- Write out meeting times instead of just verbal communication
- Begin testing earlier
- Envision more edge cases

Thank you!

Q/A